

Battery Dispatch under Forecast Uncertainty

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Abstract

Keywords: battery storage, model predictive control, probabilistic forecasting, value of information, reinforcement learning, demand charges

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Nomenclature

Abbreviations

MPC	model predictive control
PV	photovoltaic
RL	reinforcement learning
SoC	state of charge
SESOI	smallest effect size of interest
P1–P4	rule-based, point-forecast MPC, probabilistic-forecast MPC, RL policy
P2m, P2n	back-off MPC, fitted-model MPC
P3s, P3r	scenario-based and robust MPC

Symbols

t, H	hour, MPC horizon
L_t, PV_t, n_t	load, PV output, net load $L_t - PV_t$
b_t, g_t	battery power (charging positive), grid exchange (import positive)
p_t	day-ahead price
$\bar{P}, \bar{E}$	subscribed import power, export cap
ρ	overrun penalty above $\bar{P}$
m	import back-off margin of P2m
$D(b)$	degradation cost of a battery trajectory
$C_\pi(d)$	daily cost of policy π on day d
R_π	regret of policy π against the perfect-foresight oracle
V_π	share of the perfect-foresight value captured by policy π
λ	forecast-error scaling (skill)
β	weight of the CVaR term in scenario MPC

1. Introduction

2. Related work

3. Site, dispatch problem and metrics

3.1. Site and tariff

3.2. Dispatch problem

3.3. Cost, oracle and value captured

4. Forecasts and dispatch policies

4.1. Point and probabilistic forecasts

4.2. Policies

4.3. Plant variants

5. Experimental design

5.1. Data and periods

5.2. Hypotheses and tests

5.3. Inference and power

6. Results

6.1. Where the value of forecast information comes from

6.2. Probabilistic forecasts against a tuned safety margin

6.3. Value of probabilistic forecasts

6.4. Value of forecast skill

6.5. Reinforcement learning against model predictive control

7. Discussion

8. Limitations

9. Conclusion

Data availability

The code, the experiment configurations, the provenance register and the result files from which every number in this article is generated are held in the author’s project repository. They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). The input data are public (day-ahead prices, national electricity data and weather data); their sources and access dates are given in Section 5.1.

CRedit authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Software, Investigation, Validation, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

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Appendix A. Proof of Proposition P1

References